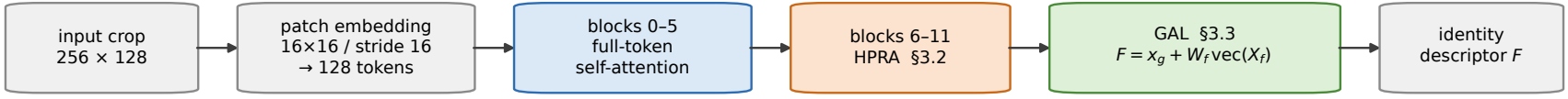
Forward pass

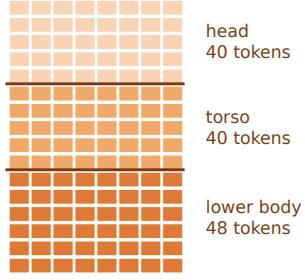
ViT-Base · 256×128 crop · 16×16 patches → 16×8 = 128 tokens · P = 3 bands



one descriptor per crop

(a) EASA — body-prior anchoring

parameter-free · no keypoints · no masks

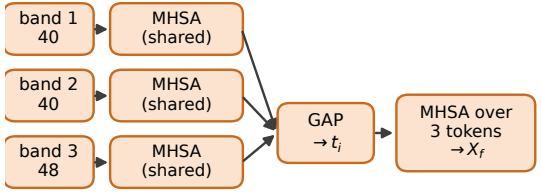


$N_i = \lfloor N_y/P \rfloor N_x, \;\; N_P = N - (P-1)N_p$

16 rows → 5 / 5 / 6 rows → 40 / 40 / 48 tokens

(b) HPRA — part-relation attention

level 1 inside the bands · level 2 across them

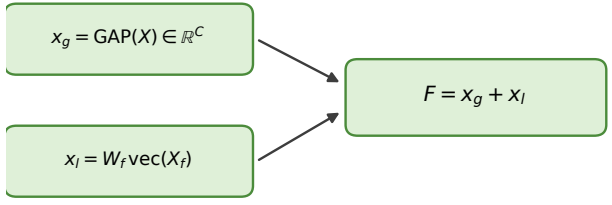


level 1: $\mathcal{O}(N^2C/P)$

level 2: $\mathcal{O}(P^2C)$, $P = 3$

(c) GAL — additive fusion

no channel expansion · no concat + FC



$W_f \in \mathbb{R}^{PC \times C}$: 1.8 M parameters

descriptor keeps the backbone

channel dimension $C = 768$